# Lecture 2. Kinematic and Dynamic Models

# Kinematic and Dynamic Models

- Vehicles (this time)
- Manipulators (next time)

# Remark

- In the context of control theory, all models describing the evolution of a system in time are dynamic models.
- In the context of robotics, there is a difference between Kinematics and Dynamics
  - Kinematics do not consider forces
  - Dynamics consider forces

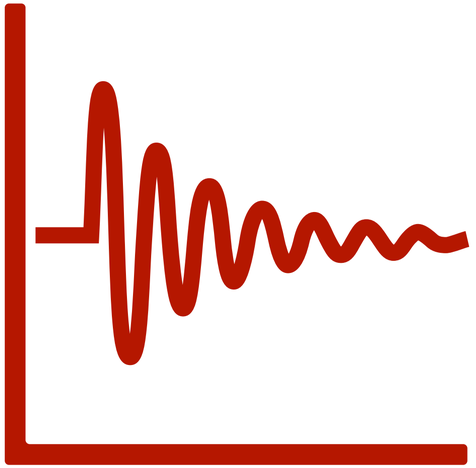
# Remark

- Continuous time (CT) models vs Discrete time (DT) models
- We may turn CT models into DT models

$$\dot{x} = f(x, u)$$

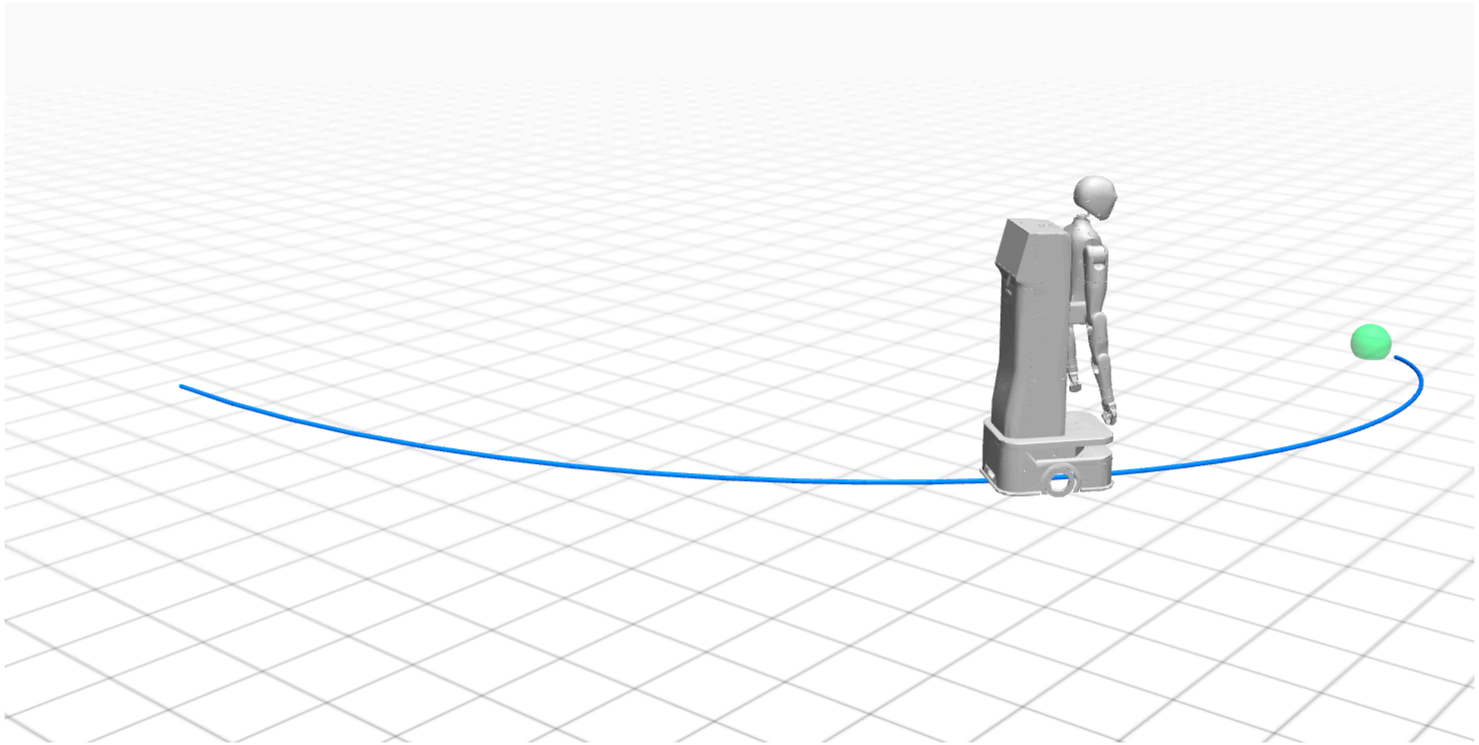
Euler Method

$$x_{k+1} = x_k + f(x_k, u_k)\Delta t$$



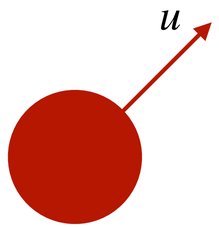
# Vehicle Models

- Single integrator
- Double integrator
- Dubins car
- Unicycle - 3 state
- Unicycle - 4 state
- Dynamic Bicycle



# Single Integrator

- State $x \in \mathbb{R}^2$: position
- Control $u \in \mathbb{R}^2$: velocity
- Dynamic equation
  - $\dot{x} = u$
  - $x_{k+1} = x_k + \Delta t \; u_k$



# Single Integrator

Advantage

Simple

Disadvantage

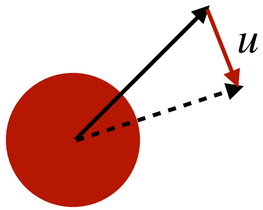
Physically not achievable (inertia and constraints not considered)

* Can you move a vehicle at $+\infty$ speed and then immediately move it at $-\infty$ speed?

* Can you immediately change the heading of a vehicle?

# Double Integrator

- State $x \in \mathbb{R}^4$: position and velocity
- Control $u \in \mathbb{R}^2$: acceleration
- Dynamic equation
  - $\dot{x} = \begin{bmatrix} 0_2 & I_2 \\ 0_2 & 0_2 \end{bmatrix} x + \begin{bmatrix} 0_2 \\ I_2 \end{bmatrix} u$
  - $x_{k+1} = \underbrace{\begin{bmatrix} I_2 & \Delta t I_2 \\ 0_2 & I_2 \end{bmatrix}}_{\text{Inertia}} x_k + \begin{bmatrix} 0_2 \\ \Delta t I_2 \end{bmatrix} u_k$




**Zero-order-hold (ZOH)**

$$x_{k+1} = \begin{bmatrix} I_2 & \Delta t I_2 \\ 0_2 & I_2 \end{bmatrix} x_k + \begin{bmatrix} \frac{\Delta t^2}{2} I_2 \\ \Delta t I_2 \end{bmatrix} u_k$$

# Double Integrator

Advantage

Relatively Simple

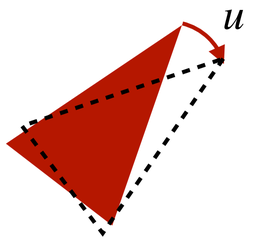
Inertia Considered

Disadvantage

Vehicle heading is not explicitly modeled

# Dubins Car

- State $x \in \mathbb{R}^3$: position and heading $x = \begin{bmatrix} x(1) \\ x(2) \\ x(3) \end{bmatrix}$
- Control $u \in \mathbb{R}$: turning rate
- Velocity $v \in \mathbb{R}$ is constant
- Dynamic equation




$$\dot{x} = \begin{bmatrix} v\cos x(3) \\ v\sin x(3) \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$x_{k+1} = x_k + \begin{bmatrix} v\cos x_k(3)\Delta t \\ v\sin x_k(3)\Delta t \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \Delta t \end{bmatrix} u_k$$

# Dubins Car

Advantage

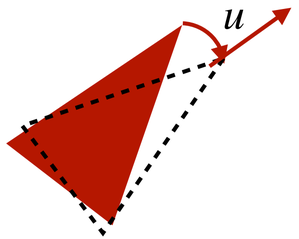
Heading considered

Disadvantage

Velocity does not change

# Unicycle - 3 State

- State $x \in \mathbb{R}^3$: position and heading $x = \begin{bmatrix} x(1) \\ x(2) \\ x(3) \end{bmatrix}$
- Control $u \in \mathbb{R}^2$: longitudinal velocity and turning rate
- Dynamic equation




$$\dot{x} = \begin{bmatrix} \cos x(3) & 0 \\ \sin x(3) & 0 \\ 0 & 1 \end{bmatrix} u$$

$$x_{k+1} = x_k + \begin{bmatrix} \cos x_k(3)\Delta t & 0 \\ \sin x_k(3)\Delta t & 0 \\ 0 & \Delta t \end{bmatrix} u_k$$

# Unicycle 3 state

Advantage

Velocity changable

Heading considered

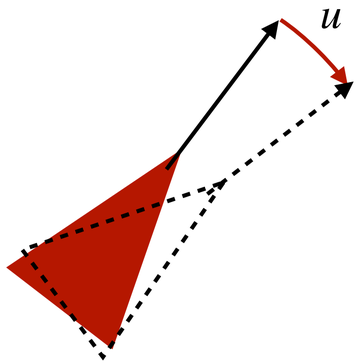
Disadvantage

Inertia not fully considered

* Systems usually cannot tolerate big jumps in velocity

# Unicycle - 4 State

- State $x \in \mathbb{R}^4$: position, longitudinal velocity, and heading
- Control $u \in \mathbb{R}^2$: longitudinal acceleration, and turning rate
- Dynamic equation




$$x = \begin{bmatrix} x(1) \\ x(2) \\ x(3) \\ x(4) \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} x(3)\cos x(4) \\ x(3)\sin x(4) \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u$$

$$x_{k+1} = x_k + \begin{bmatrix} x(3)\cos x(4)\Delta t \\ x(3)\sin x(4)\Delta t \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix} u_k$$

# Unicycle 4 state

Advantage

Inertia Considered

Heading considered

Disadvantage

More advanced dynamics not considered, e.g., drift

# Bicycle Dynamic Model

- State $x \in \mathbb{R}^6$: position, longitudinal velocity, lateral velocity, yaw rate, yaw angle
  $x = [p_x \ p_y \ v_x \ v_y \ r \ \psi]^\top$
- Control $u \in \mathbb{R}^2$: longitudinal force, and front wheel steering angle
  $u = [F_x \ \delta_f]^\top$
- Dynamic equation

$$\dot{p}_x = v_x \cos\psi - v_y \sin\psi$$

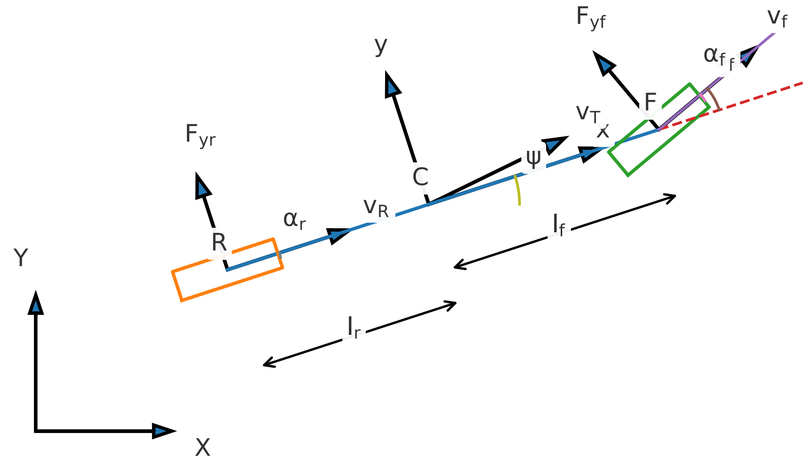
$$\dot{p}_y = v_x \sin\psi + v_y \cos\psi$$

$$m(\dot{v}_x - r v_y) = F_{xf}\cos\delta_f - F_{yf}\sin\delta_f + F_{xr}$$

$$m(\dot{v}_y + r v_x) = F_{xf}\sin\delta_f + F_{yf}\cos\delta_f + F_{yr}$$

$$I_z\dot{r} = l_f\left(F_{xf}\sin\delta_f + F_{yf}\cos\delta_f\right) - l_r F_{yr}$$

$$\dot{\psi} = r$$




$$F_{xf} = \beta F_x, \qquad F_{xr} = (1-\beta) F_x, \qquad 0 \le \beta \le 1$$

$$F_{yf} = -C_f\left[\arctan\left(\frac{v_y + l_f r}{\max(v_x, \varepsilon)}\right) - \delta_f\right]$$

$$F_{yr} = -C_r \arctan\left(\frac{v_y - l_r r}{\max(v_x, \varepsilon)}\right)$$

# Bicycle Model

Advantage

Inertia Considered

Geometry considered

Drift considered

Disadvantage

Highly nonlinear

# Kinematic Vehicle Models

| Model | State | Control | Continuous Time Dynamics | Discrete Time Dynamics |
|---|---|---|---|---|
| Single Integrator | | | | |
| Double Integrator | | | | |
| Dubins | | | | |
| Unicycle3 | | | | |
| Unicycle4 | | | | |
| Dynamic Bicycle | | | | |

# Remarks

- There are other more advanced models:
  - Four-wheel model: more like a passenger vehicle
- When do we use these models?
  - To control the ego vehicle
  - To predict the motion of other vehicles

# Remarks

- What if the model is not accurate?
  - If we are controlling a real hardware system, there will always be **model mismatch**.
  - There are control techniques to compensate the model mismatch.*
  - The purpose to study these simple models is to derive control strategies that can be applied to more complex models.

# Remarks

- How to choose among different models?
  - We need to pick a model that is most suitable for the task.
    - For highway driving, it is more important to consider inertia than constraints. Then we can use double integrator.
    - For low-speed driving (e.g., in parking lots), it is more important to consider constraints than inertia. Then we can use unicycle (3 state).

# Remarks

- Linear Models
  - $\dot{x} = Ax + Bu$
  - $x_{k+1} = Ax_k + Bu_k$
- Among all models we discussed, which of them are linear?

# Remarks

- Although not all models are linear, one thing in common for all kinematic models (excluding the dynamic bicycle model):
  - The models are linear in control
    - $\dot{x} = f(x) + g(x)u$
    - $x_{k+1} = f(x_k) + g(x_k)u_k$
  - These models are called **control-affine dynamics**

# Model Linearization

$x_{k+1} = f(x_k, u_k)$ **Reference** $(\bar{x}_k, \bar{u}_k)$

$$\delta x_k := x_k - \bar{x}_k, \qquad \delta u_k := u_k - \bar{u}_k$$

$$\delta x_{k+1} \approx A_k \delta x_k + B_k \delta u_k + d_k$$

$$A_k = \frac{\partial f}{\partial x}\Big|_{(\bar{x}_k, \bar{u}_k)}, \quad B_k = \frac{\partial f}{\partial u}\Big|_{(\bar{x}_k, \bar{u}_k)}, \quad d_k = f(\bar{x}_k, \bar{u}_k) - \bar{x}_{k+1}$$

**Error in the reference trajectory**

**Example: Discrete Time Unicycle 3-State**

$$x_k = [p_k^x; p_k^y; \theta_k] \qquad u_k = [v_k; w_k]$$

$$x_{k+1} = x_k + \begin{bmatrix} \cos\theta_k \Delta t & 0 \\ \sin\theta_k \Delta t & 0 \\ 0 & \Delta t \end{bmatrix} u_k$$

$$A_k = \begin{bmatrix} 1 & 0 & -\Delta t\, \bar{v}_k \sin\bar{\theta}_k \\ 0 & 1 & \Delta t\, \bar{v}_k \cos\bar{\theta}_k \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_k = \begin{bmatrix} \Delta t \cos\bar{\theta}_k & 0 \\ \Delta t \sin\bar{\theta}_k & 0 \\ 0 & \Delta t \end{bmatrix}$$

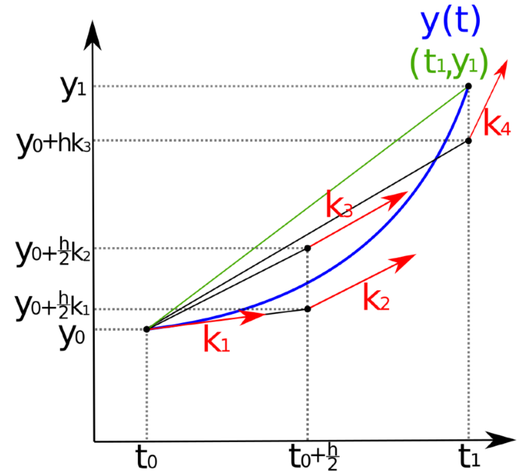
# Simulation

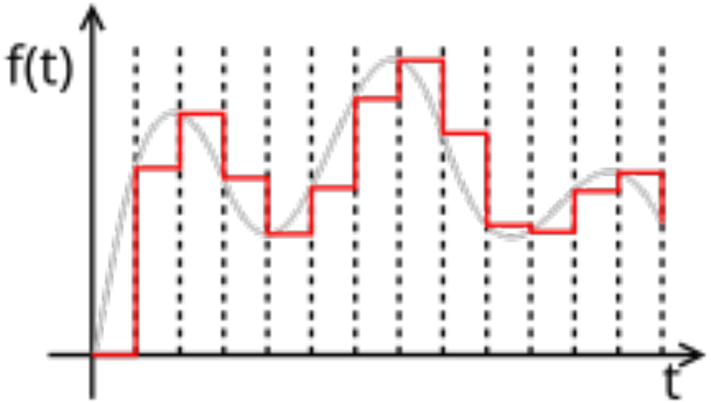
- Simulation
  - Matlab: standalone, control oriented
  - Python (New in 2026): integration with more robotics tools
    - Tutorial next lecture

**https://github.com/intelligent-control-lab/16-714**

# Continuous Time to Discrete Time

- Euler
  - $x_{k+1} = x_k + f(x_k, u_k)\Delta t$
- ZOH
  - $x_{k+1} = x_k + \int_{t=k}^{k+1} f(x_t, u_t)dt$
- RK4

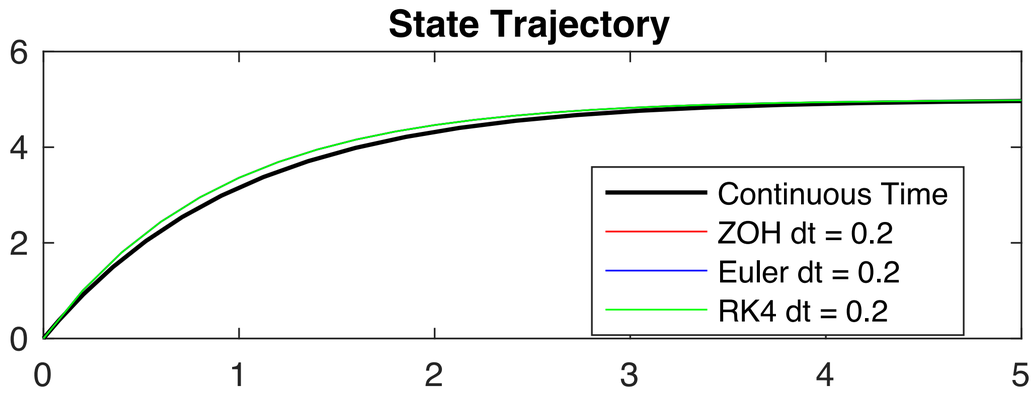


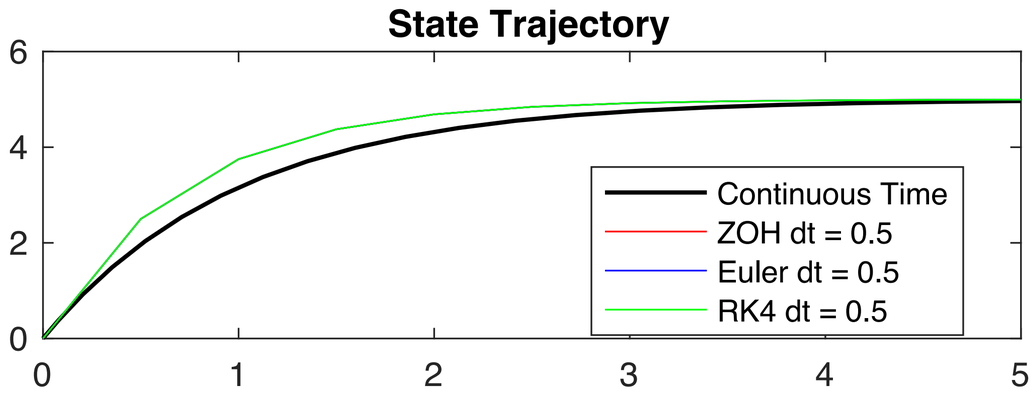



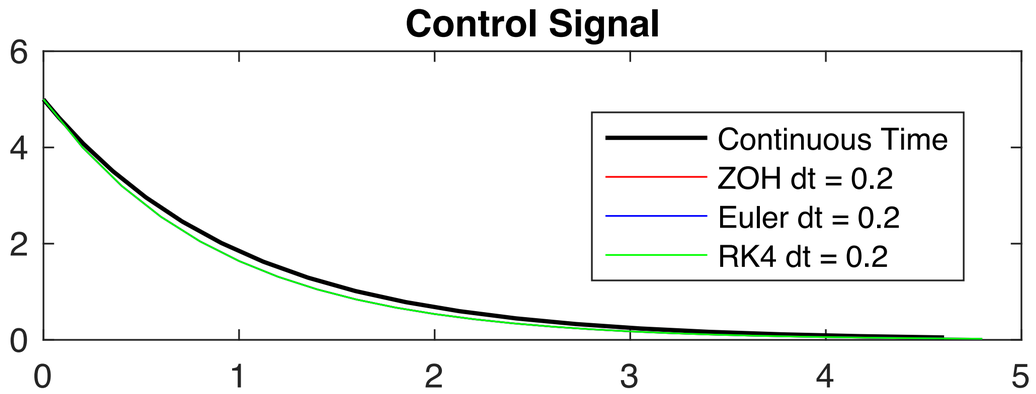

https://lpsa.swarthmore.edu/NumInt/NumIntFourth.html

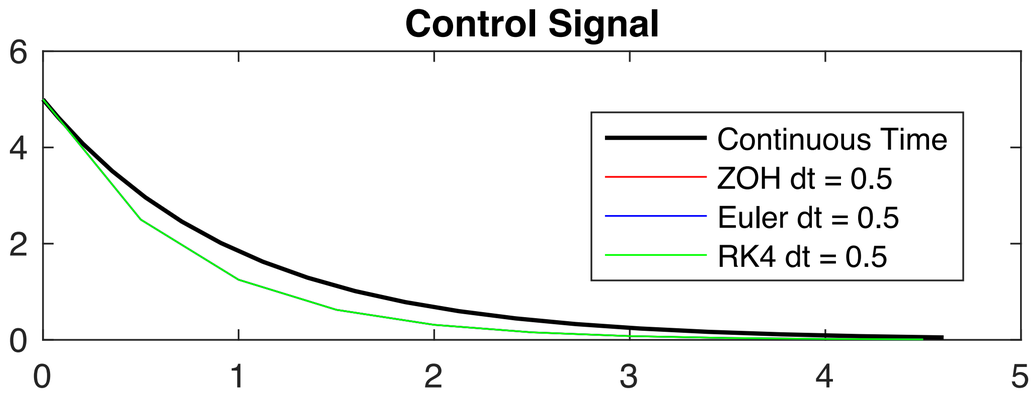
# Mismatch Between CT & DT

*** There is no difference among the three discretization methods**



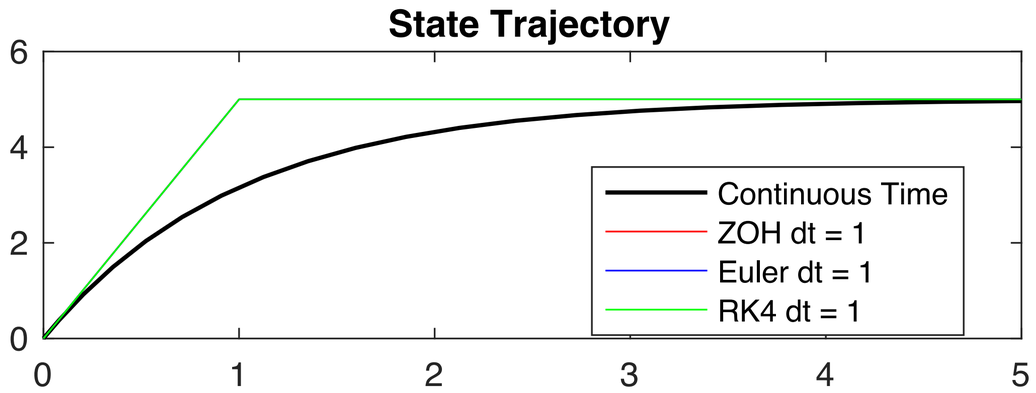



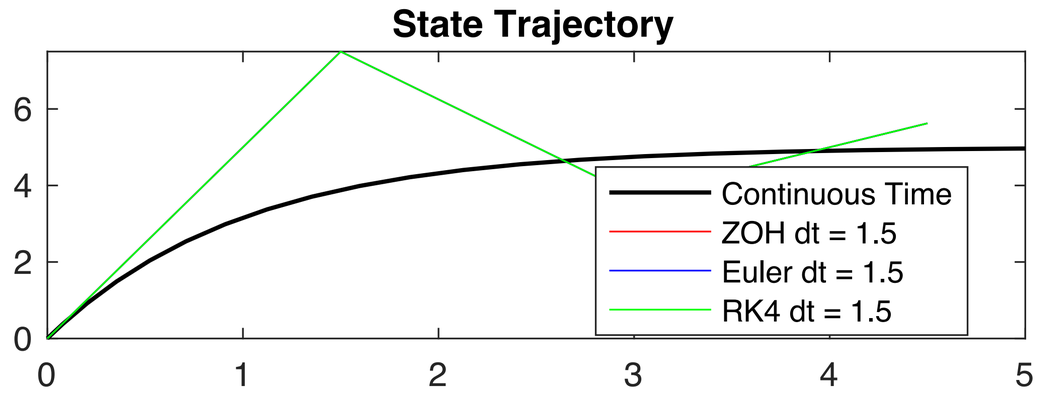



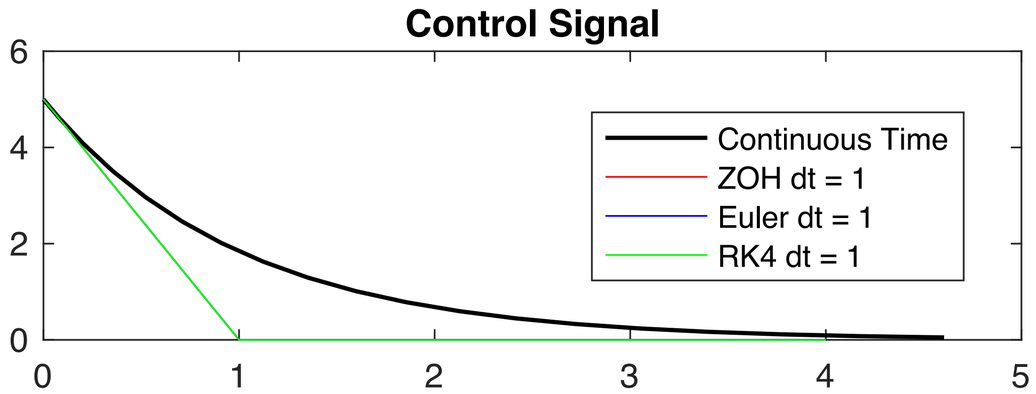



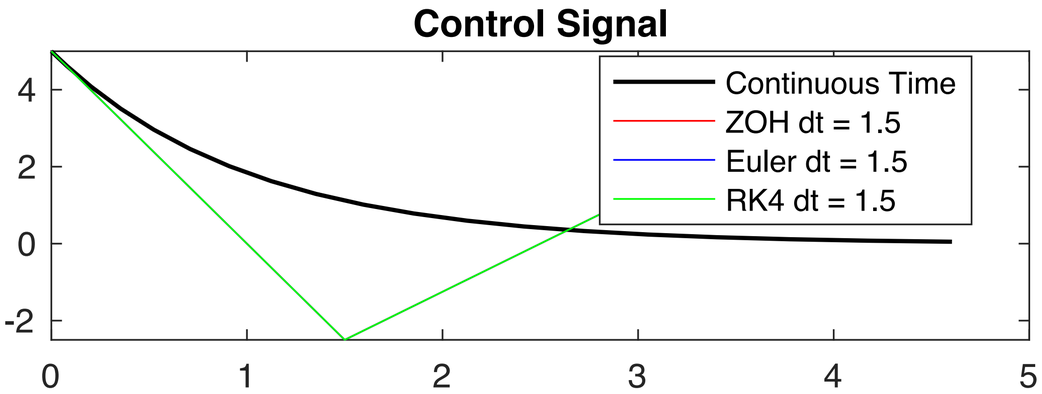

```
roll_out("single_integrator", @(x,t) 5 - x, 0, simCT);
roll_out("single_integrator", @(x,t) 5 - x, 0, simDT_ZOH);
roll_out("single_integrator", @(x,t) 5 - x, 0, simDT_Euler);
roll_out("single_integrator", @(x,t) 5 - x, 0, simDT_RK4);
```

# Mismatch Between CT & DT




```
roll_out("single_integrator", @(x,t) 5 - x, 0, simCT);
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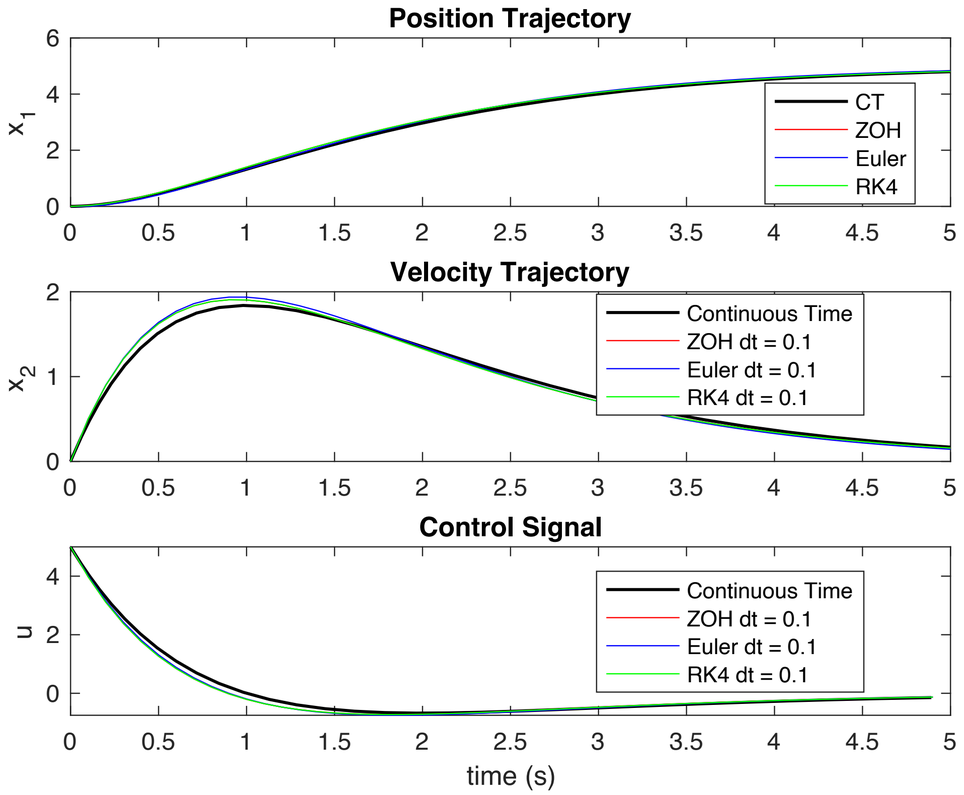
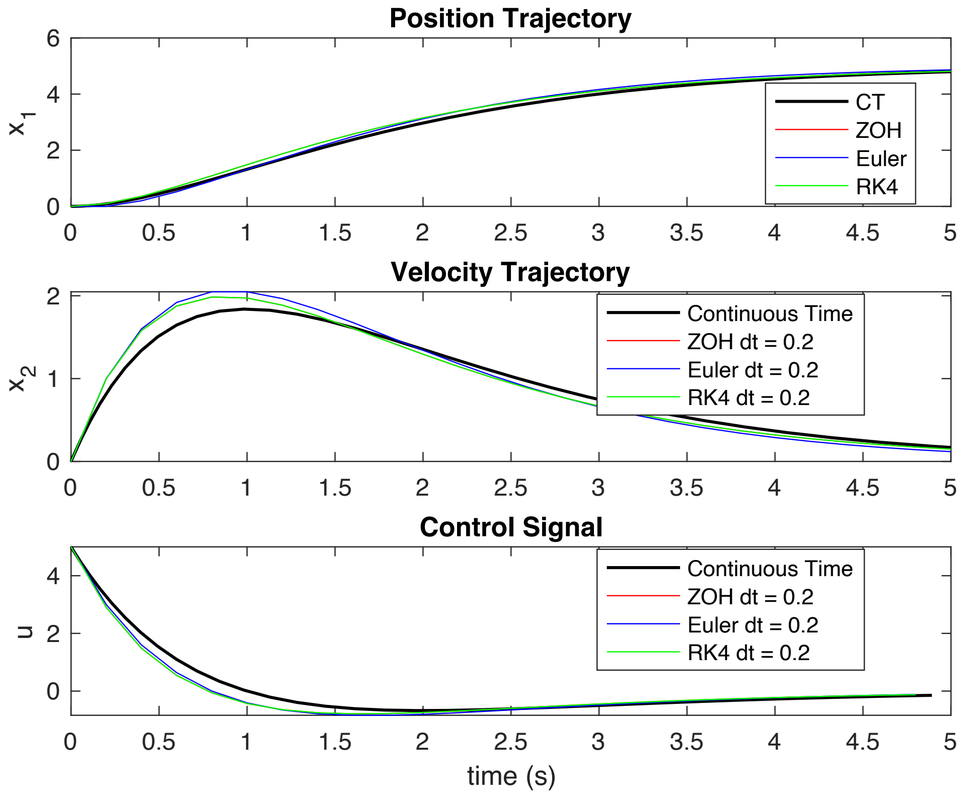
# Mismatch Between CT & DT

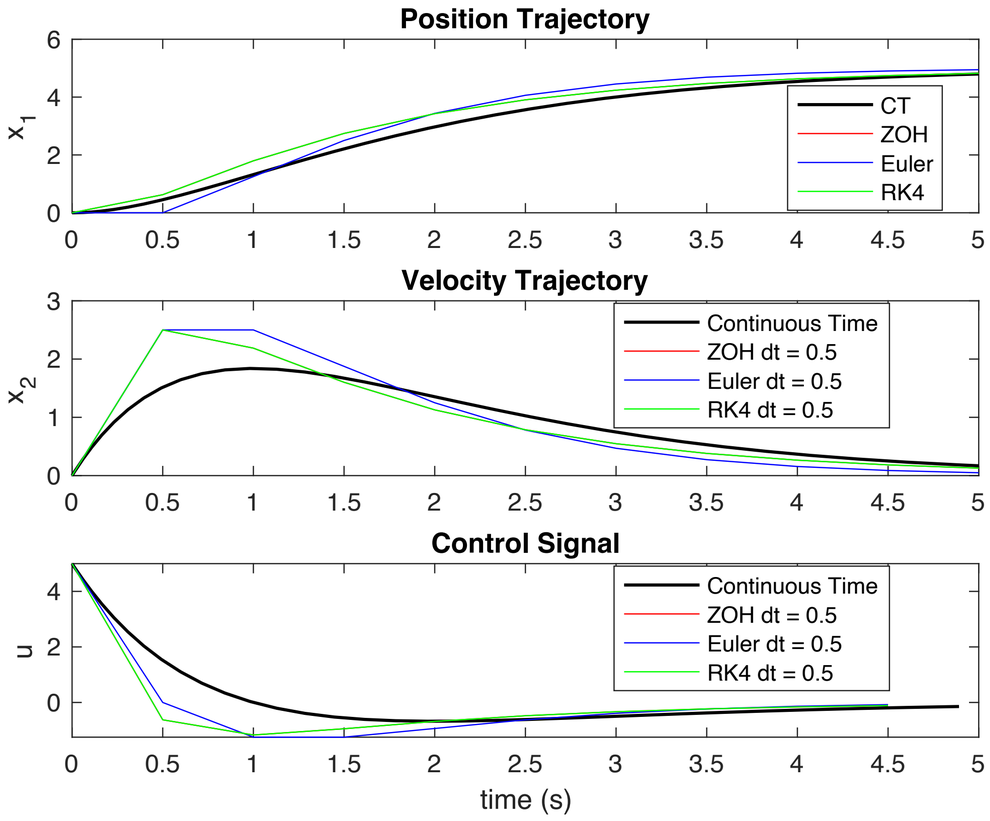


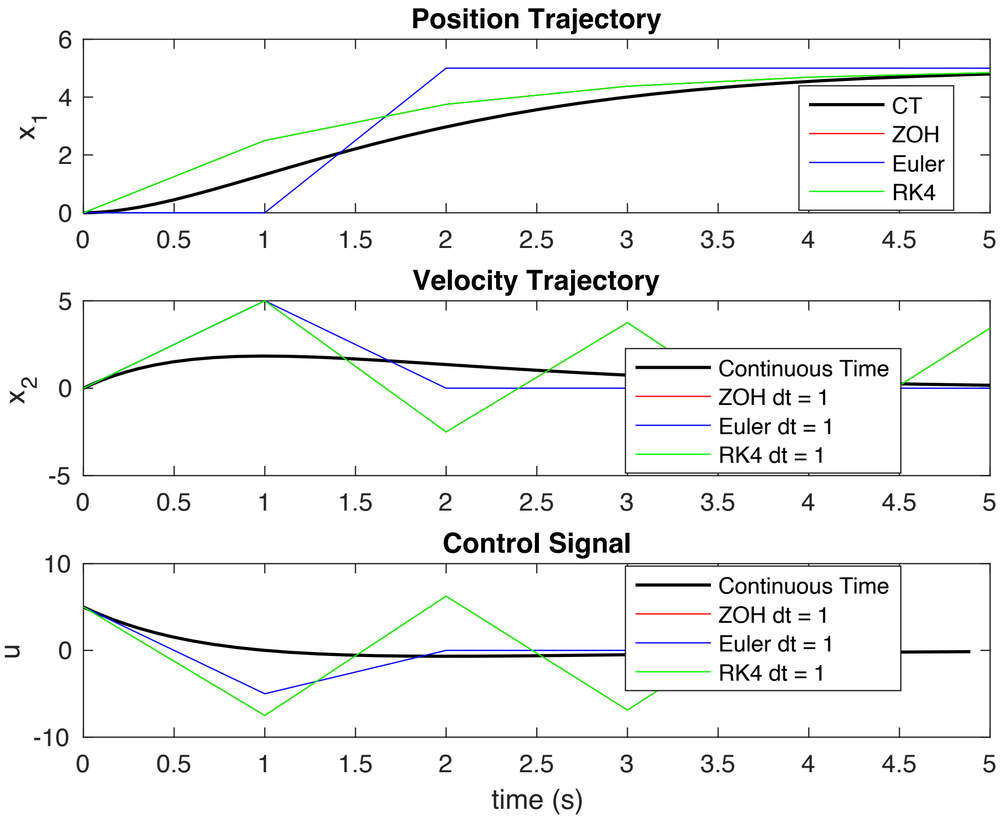




```
roll_out("double_integrator", @(x,t) 5 - x(1) - 2*x(2), [0;0], simCT);
roll_out("double_integrator", @(x,t) 5 - x(1) - 2*x(2), [0;0], simDT_ZOH);
roll_out("double_integrator", @(x,t) 5 - x(1) - 2*x(2), [0;0], simDT_Euler);
roll_out("double_integrator", @(x,t) 5 - x(1) - 2*x(2), [0;0], simDT_RK4);
```

*** State feedback control**

# Mismatch Between CT & DT

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```
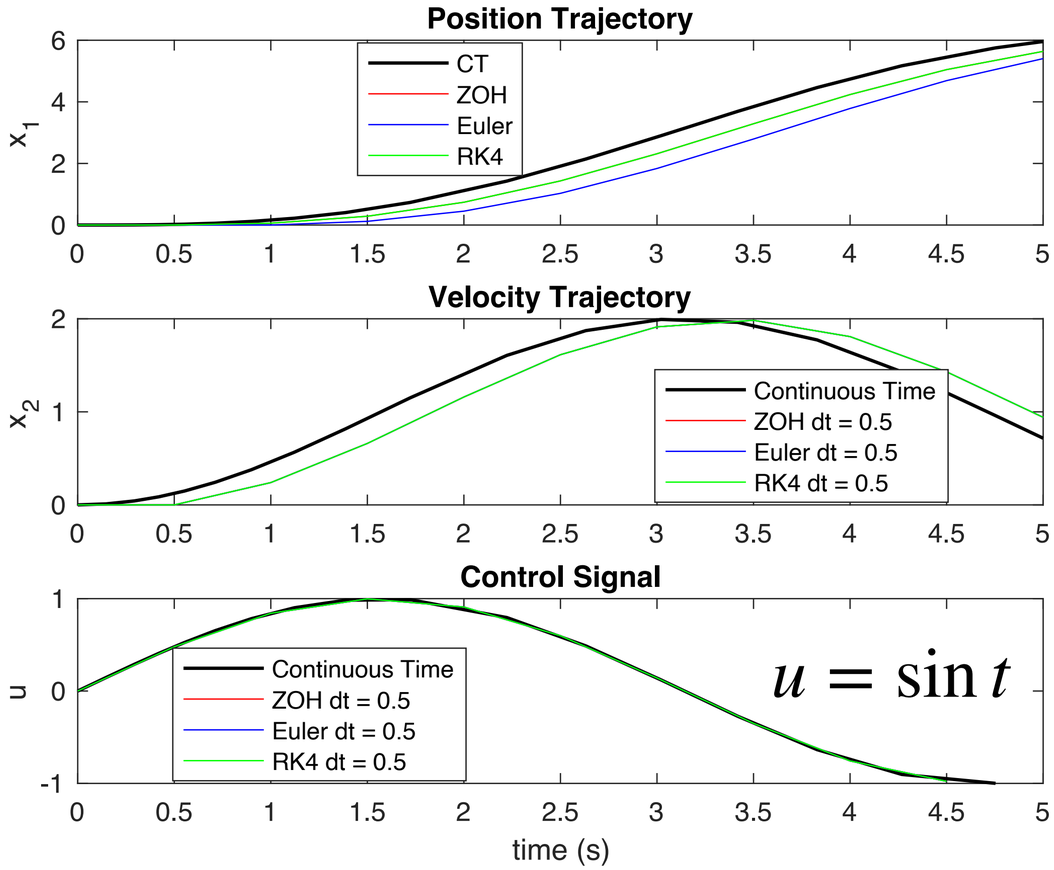

# Mismatch Between CT & DT




```
roll_out("double_integrator", @(x,t) sin(t), [0;0], simCT);
roll_out("double_integrator", @(x,t) sin(t), [0;0], simDT_ZOH);
roll_out("double_integrator", @(x,t) sin(t), [0;0], simDT_Euler);
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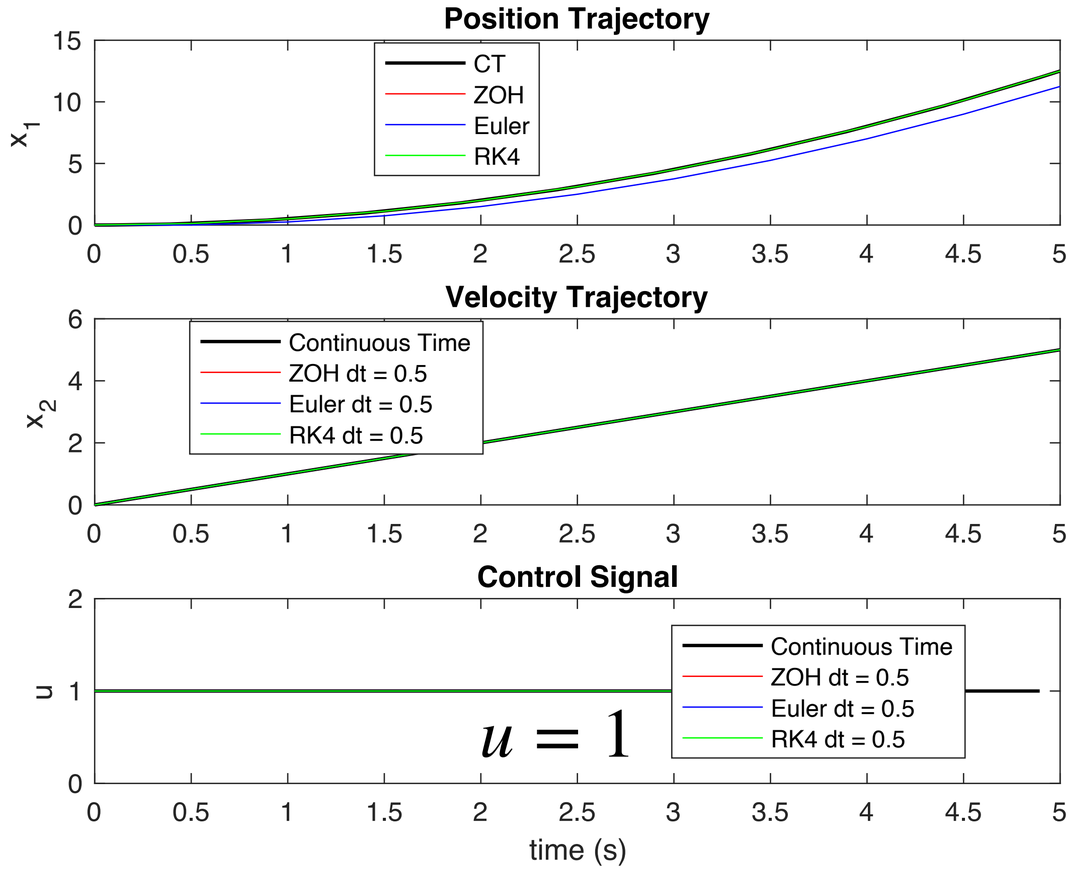


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roll_out("double_integrator", @(x,t) 1, [0;0], simCT);
roll_out("double_integrator", @(x,t) 1, [0;0], simDT_ZOH);
roll_out("double_integrator", @(x,t) 1, [0;0], simDT_Euler);
roll_out("double_integrator", @(x,t) 1, [0;0], simDT_RK4);
```
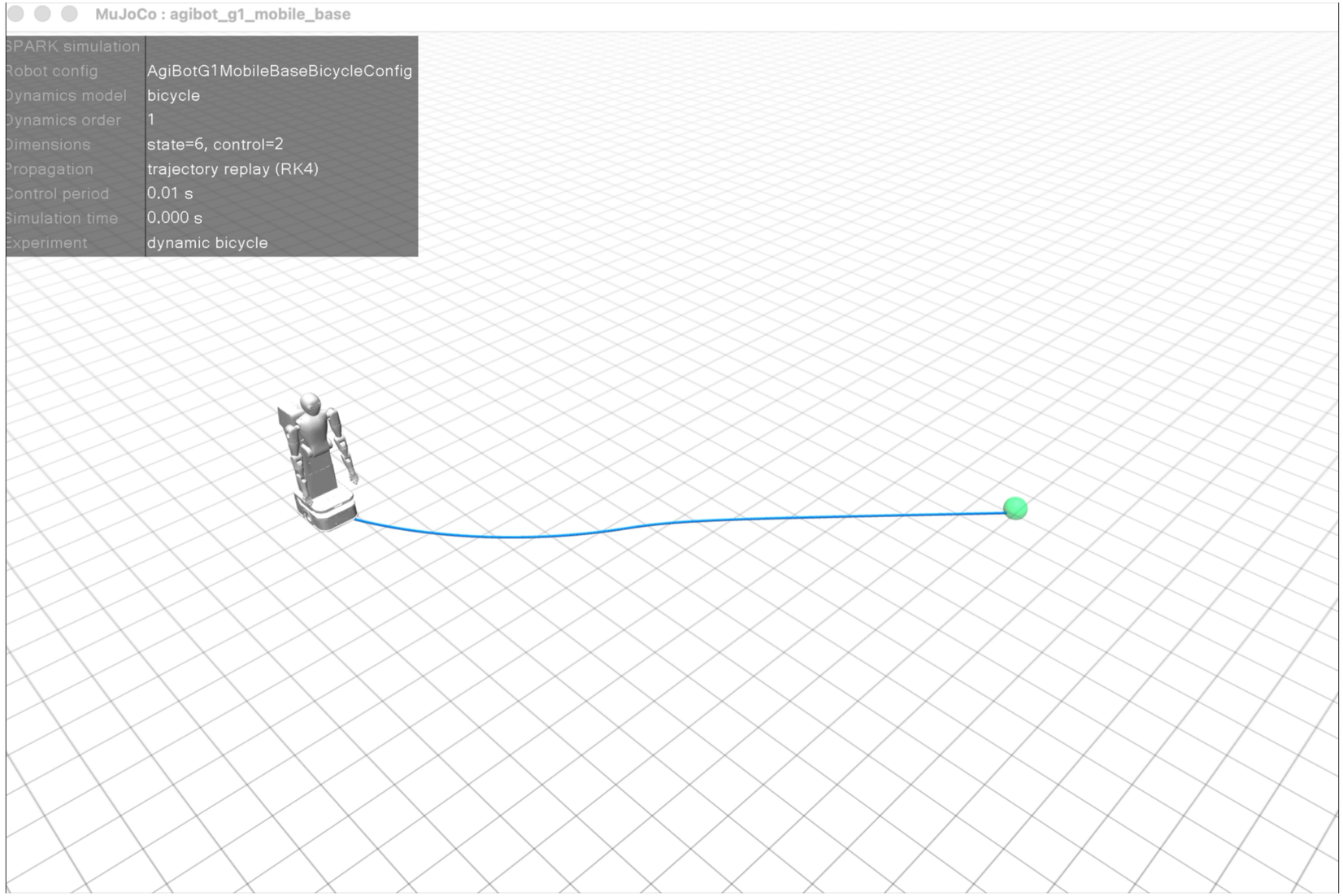

# Remark

- Under the same control law, the gaps between CT trajectories and DT trajectories could become bigger with larger sampling time, especially when the control signal is state-dependent or time-dependent

- In practice, people design different controllers for DT systems to account for the sampling time.

# Vehicle Trajectory Visualization



# Next Time

- Models for Manipulators